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# EVERYTHING I KNOW ABOUT TIME SERIES



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# *Difference Equations*

The whole point of time series analysis is to uncover and describe the statistical properties and structure of a process across time. Certain processes may have a lot of structure, so much that advanced non-linear methods such as those enabled by machine learning are needed. Others have very little, and can be adequately described by a very simple model. As we are always worried about overfitting, and there is a general philosophy that parsimony is a good thing, we begin by studying the simplest of processes.

Assume we have the deterministic process

$$y_t = \phi y_{t-1} + w_t \tag{1}$$

Test